Geometric Transformation of Financial Time Series: A Vector Approach with Logarithmic Returns for Price Direction Prediction

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Abstract

This paper presents a novel geometric transformation for financial time series based on logarithmic returns, where each price movement is represented as a two-dimensional vector with a fixed origin at $(0, 0)$. The main contribution lies in the interpretation of the entire price dynamics as a **directional compass**: a polar coordinate system where the angle indicates the direction of the movement and the radial magnitude represents the normalized intensity. The fundamental properties of this transformation are formally proven: Lipschitz continuity, sign preservation, scale invariance, and uniform boundedness. A rigorous bridge with Takens' embedding theorem is established. The methodology is empirically validated with 2155 observations of natural gas (NG=F, 2017-2026) and 2155 observations of Brent crude oil (BZ=F, 2017-2026), using AR, MLP, LSTM, and ARMA-GARCH/EGARCH/GJR-GARCH models. Results show that both energy assets exhibit stationary angular series with bounded angles. For natural gas, the MLP model achieves the lowest MAE (2.1270°) and directional accuracy (51.40%). For Brent, the AR(1) and LSTM models achieve superior performance with MAE of 0.9455° and directional accuracy of 51.86%, consistently outperforming random levels. The comparative analysis reveals that Brent exhibits lower angular volatility (standard deviation 1.46° vs 2.60°), suggesting more predictable directional dynamics. The relevance of the approach in the current context of geopolitical tensions is analyzed, demonstrating that the proposed turbulence index acts as a leading indicator of crises in energy markets.

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Prologue: The Compass as a Foundational Metaphor

> "The compass did not create the sea, but it made it navigable. It did not calm the storms, but it taught how to find one's way through them."
>
> *Anonymous*

In ancient China, during the Han Dynasty (206 BC – 220 AD), scholars discovered that a lodestone — magnetite — suspended in the air always pointed south. They did not understand the physics of terrestrial magnetism, but they grasped something deeper: **direction is information**. With that knowledge, they transformed navigation. Before the compass, sailors navigated by the stars, the coast, and intuition. When the sky clouded over or the coast disappeared, they were left adrift. The compass did not create new seas nor calm the storms, but it offered something more valuable: a **stable point of reference** amidst uncertainty.

Financial markets resemble those uncertain seas. Every day, millions of transactions generate erratic movements that investors try to interpret. Prices rise and fall. Volatility increases and decreases. Crises, like storms, appear without warning — a war here, a default there, an unexpected pandemic. In this apparent chaos, financial analysts have sought their own "compass" for centuries: mathematical models to navigate price turbulence.

Traditional models — ARIMA, GARCH, neural networks — are like stars: powerful when the sky is clear, but useless when the geopolitical storm darkens the horizon. Their main limitation is that they operate directly on returns, a variable with no upper bound: a 5%, 15%, 50% return... where is the limit? In a crisis, returns can explode numerically, making models unstable.

This work proposes a financial compass. The idea is as simple as it is profound: instead of looking at returns, we look at the **direction** they indicate. Each price movement becomes an angle: 0° if the price goes up, 180° if it goes down. But unlike the stars — which hide behind clouds — this compass works even in the fiercest storm because its angles are always bounded between $-90°$ and $90°$. A 10% return and a 50% return produce different angles, but both fit within the same numerical range. The compass does not eliminate market volatility, but it offers a stable framework for studying it.

Like the ancient Chinese, we do not need to understand all the forces that move prices to build a tool that guides us. We need, however, to dare to look at the market from another perspective. This is my proposal: a directional compass to navigate the financial seas.

1 Introduction

1.1 Motivation and Context

The analysis of financial time series constitutes one of the most significant challenges in contemporary econometrics. The stochastic, non-stationary, and heteroscedastic nature of asset prices has motivated the development of numerous predictive approaches. Tra-

ditionally, the literature has focused on parametric models such as ARIMA [4], GARCH [5, 6] and their asymmetric extensions — EGARCH [7] and GJR-GARCH [8] — as well as non-parametric approaches based on machine learning [13].

However, there is a fundamental disconnect between the geometric nature of price movements and conventional mathematical representations. Prices are inherently directional, but traditional models treat returns as isolated scalar variables, losing information about the underlying geometric structure of the trajectories. This limitation motivates our proposal: to represent price movements as vectors in a geometric space where the direction explicitly captures the market trend.

1.2 The Novelty of the Approach: A Compass for the Market

The fundamental contribution of this work is the reinterpretation of financial dynamics as a directional navigation system. I propose that each price movement can be visualized as a compass needle pointing in a specific direction:

- **Angle** 0° **(East)**: Price increase (positive return)
- **Angle** 90° **(North)**: Zero return (price unchanged)
- **Angle** 180° **(West)**: Price decrease (negative return)

This analogy is not merely aesthetic: the compass provides an intuitive, geometric representation where direction indicates the trend and distance from the center encodes the normalized magnitude of the movement. Unlike traditional price charts, which require training to interpret, the directional compass offers immediate reading: a concentration of angles in the positive hemisphere indicates a bullish market; high angular dispersion signals uncertainty or regime transition.

1.3 Contributions of this Work

The main contributions of this paper are:

1. A vector representation for financial series based on logarithmic returns, with formal proof of its properties.
2. The introduction of the concept of a **financial directional compass** as a visualization tool and geometric interpretation.
3. An application of Takens' embedding theorem to the angular space.
4. A predictive methodology that combines circular statistics, GARCH models with leverage effects, and deep architectures (LSTM, MLP).
5. Rigorous empirical validation with 2155 observations of natural gas (NG=F, 2017-2026) and 2155 observations of Brent crude oil (BZ=F, 2017-2026), two high-volatility assets representative of energy markets.
6. A comparative analysis between natural gas and Brent, revealing distinct angular signatures for each asset.
7. An analysis of geopolitical relevance applied to the current conflict between the United States, Israel, and Iran (April 2026), demonstrating that the turbulence index $\mathcal{T}_t$ acts as a leading indicator of crises in energy markets.

2 Theoretical Framework

2.1 Logarithmic Returns

Definition 2.1 (Logarithmic Return)**.** *For a price series $\{P_t\}_{t=0}^{T} \subset \mathbb{R}_+$, we define the logarithmic return at time t as:*

$$r_t = \ln\left(\frac{P_{t+1}}{P_t}\right) = \ln P_{t+1} - \ln P_t, \qquad t = 0, 1, \ldots, T-1 \tag{1}$$

Theorem 2.1 (Properties of Logarithmic Returns)**.** *Logarithmic returns satisfy:*

1. ***Temporal additivity****: $r_{t,t+k} = \sum_{i=0}^{k-1} r_{t+i}$*
2. ***Weak stationarity****: Under the efficient market hypothesis, $\mathbb{E}[r_t] = \mu$ and $Var(r_t) = \sigma^2$ are constant*
3. ***Scale invariance****: Invariant under transformations $P_t \mapsto \lambda P_t$*

2.2 Takens' Embedding Theorem

Takens' theorem [9] establishes the conditions under which a dynamical system can be reconstructed from one-dimensional observations.

Theorem 2.2 (Takens, 1981)**.** *Let M be a compact manifold of dimension d, $\phi : M \to M$ a C^2 diffeomorphism, and $y : M \to \mathbb{R}$ a C^2 function. Then, for generic ϕ and y, the map $\Phi_{\phi,y} : M \to \mathbb{R}^{2d+1}$ defined by*

$$\Phi_{\phi,y}(x) = (y(x), y(\phi^{-1}(x)), \ldots, y(\phi^{-2d}(x))) \tag{2}$$

is an embedding of M into $\mathbb{R}^{2d+1}$.

2.3 Volatility Models with Asymmetries

2.3.1 EGARCH Model

The EGARCH (Exponential GARCH) model [7] captures leverage effects:

$$\ln(\sigma_t^2) = \omega + \sum_{i=1}^{p} \alpha_i \left|\frac{\varepsilon_{t-i}}{\sigma_{t-i}}\right| + \sum_{i=1}^{p} \gamma_i \frac{\varepsilon_{t-i}}{\sigma_{t-i}} + \sum_{j=1}^{q} \beta_j \ln(\sigma_{t-j}^2) \tag{3}$$

The coefficient γ_i measures the leverage effect: $\gamma_i < 0$ indicates that negative shocks increase volatility more than positive ones.

2.3.2 GJR-GARCH Model

The GJR-GARCH model [8] introduces asymmetry via an indicator function:

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} (\alpha_i + \gamma_i I_{t-i})\varepsilon_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2 \tag{4}$$

where $I_{t-i} = \mathbf{1}_{\{\varepsilon_{t-i}<0\}}$.

2.4 Circular Statistics

Circular statistics [17, 18] provides the necessary tools for the analysis of angular data.

Definition 2.2 (Circular Measures). *For a sample of angles $\{\theta_t\}_{t=1}^N$, we define:*

$$\textit{Circular mean: } \bar{\theta}_c = \arctan\left(\frac{\sum_{t=1}^N \sin\theta_t}{\sum_{t=1}^N \cos\theta_t}\right) \tag{5}$$

$$\textit{Circular variance: } V_c = 1 - \frac{1}{N}\sqrt{\left(\sum_{t=1}^N \cos\theta_t\right)^2 + \left(\sum_{t=1}^N \sin\theta_t\right)^2} \tag{6}$$

$$\textit{Circular standard deviation: } S_c = \sqrt{-2\ln(1-V_c)} \tag{7}$$

3 Geometric Transformation: The Directional Compass

3.1 Formal Definition

Definition 3.1 (Logarithmic Movement Vector). *For each time t, we define the movement vector:*

$$\vec{v}_t = (1, r_t) \in \mathbb{R}^2 \tag{8}$$

Definition 3.2 (Directional Angle Function). *The angular transformation $\Theta : \mathbb{R}^2 \to (-\pi/2, \pi/2)$ is defined as:*

$$\theta_t = \Theta(\vec{v}_t) = \arctan\left(\frac{r_t}{1}\right) = \arctan(r_t) \tag{9}$$

Definition 3.3 (Financial Directional Compass). *We define the **directional compass** as the mapping $\mathcal{C} : \mathbb{R}_+ \times \mathbb{R} \to [0, 2\pi) \times [0, 1]$ that assigns to each price-return pair a polar coordinate:*

$$\mathcal{C}(P_t, r_t) = (\theta_t, \rho_t), \quad \textit{where } \theta_t = \arctan(r_t), \quad \rho_t = \frac{|r_t|}{\max_s |r_s|} \tag{10}$$

The radius $\rho_t \in [0, 1]$ normalizes the magnitude of the movement relative to the historical maximum.

3.2 Fundamental Properties

Theorem 3.1 (Sign Preservation). *$sign(\theta_t) = sign(r_t)$ for all t.*

Theorem 3.2 (Uniform Boundedness). *For financial series, we have:*

$$|\theta_t| \leq \arctan(\rho) \ll \frac{\pi}{2} \tag{11}$$

Theorem 3.3 (Lipschitz Continuity). *The transformation Θ is Lipschitz continuous with constant $L = 1$:*

$$|\theta_i - \theta_j| \leq |r_i - r_j|, \qquad \forall i, j \tag{12}$$

Theorem 3.4 (Scale Invariance). *The transformation is invariant under price scaling: if $P'_t = \lambda P_t$, then $\theta'_t = \theta_t$.*

3.3 Angular Embedding

Based on Takens' theorem, we construct an angular state space.

Definition 3.4 (Angular Embedding Vector)**.** *For embedding dimension m, we define:*

$$\boldsymbol{\theta}_t^{(m)} = (\theta_t, \theta_{t-1}, \ldots, \theta_{t-m+1}) \in \mathbb{R}^m \tag{13}$$

In our implementation, the embedding dimension was set to $m = 7$ for natural gas and $m = 6$ for Brent, determined via partial autocorrelation analysis (PACF).

4 Predictive Methodology

4.1 Prediction Strategy

Prediction is performed in the angular space using the following formulation:

$$\hat{\theta}_{t+1} = f(\boldsymbol{\theta}_t^{(m)}; \boldsymbol{\phi}) \tag{14}$$

We evaluate four families of models:

1. **AR(1)**: First-order autoregressive model (linear benchmark).
2. **MLP**: Multilayer perceptron with one hidden layer and ReLU activation.
3. **LSTM**: Recurrent network with 32 units, 20% dropout, and L2 regularization.
4. **ARMA-GARCH/EGARCH/GJR-GARCH**: Econometric models with conditional volatility modeling.

4.2 Evaluation Metrics

Definition 4.1 (Mean Absolute Angular Error)**.**

$$MAE_\theta = \frac{1}{N} \sum_{t=1}^{N} |\theta_t - \hat{\theta}_t| \tag{15}$$

Definition 4.2 (Directional Accuracy)**.**

$$DirAcc = \frac{1}{N} \sum_{t=1}^{N} \mathbf{1}_{\{sign(\theta_t) = sign(\hat{\theta}_t)\}} \tag{16}$$

Definition 4.3 (Theil's U Statistic)**.**

$$U = \sqrt{\frac{\sum_{t=1}^{N} (\theta_t - \hat{\theta}_t)^2}{\sum_{t=1}^{N} \theta_t^2}} \tag{17}$$

A value $U < 1$ indicates that the model outperforms the naive forecast.

Definition 4.4 (Circular Correlation)**.**

$$\rho_c(\theta, \hat{\theta}) = \frac{\sum_{t=1}^{N} \sin(\theta_t - \bar{\theta}) \sin(\hat{\theta}_t - \bar{\hat{\theta}})}{\sqrt{\sum_{t=1}^{N} \sin^2(\theta_t - \bar{\theta}) \sum_{t=1}^{N} \sin^2(\hat{\theta}_t - \bar{\hat{\theta}})}} \tag{18}$$

4.3 Temporal Validation

To respect the sequential nature of the data, we implement a strict temporal split:

- **Training**: 64% of samples
- **Validation**: 16% of samples
- **Test**: 20% of samples

5 Experimental Results: Natural Gas (NG=F)

5.1 Experimental Setup

The study was conducted on the NG=F asset (Natural Gas, NYMEX) over the period from September 2017 to April 2026, totaling 2155 daily observations. Table 1 summarizes the experimental setup.

Table 1: Experimental Setup for Natural Gas (NG=F)

Parameter	**Value**
Asset	NG=F (Natural Gas, NYMEX)
Period	2017-09-10 to 2026-04-03
Frequency	Daily
Number of observations	2155
Calculated returns	2154
Return range	$[-0.6440, 0.3817]$
Angle range	$[-32.78°, 20.89°]$
Embedding window	$m = 7$ (fixed)
Data split	1373 training, 344 validation, 430 test
Models evaluated	AR(1), MLP, LSTM, ARMA-GARCH, ARMA-EGARCH, ARMA-GJR-GARCH

5.2 Descriptive Statistics of the Angular Transformation

Table 2 presents the descriptive statistics of the angular series resulting from applying the transformation $\theta_t = \arctan(r_t)$.

Table 2: Descriptive Statistics of Directional Angles (NG=F)

Metric	**Value**
Arithmetic mean	0.0001°
Standard deviation	2.6014°
Minimum	−32.7805°
Maximum	20.8932°
Circular mean	0.0007°
Circular variance	0.001027
Circular standard deviation	0.0453 rad
95th percentile ($\lvert\theta\rvert$)	4.8833°

The mean close to zero confirms the absence of systematic directional bias in the analyzed period. The circular variance of 0.001027 is higher than that observed in lower-volatility assets, reflecting the more erratic nature of natural gas.

Remark 5.1 (Stationarity)**.** *The augmented Dickey-Fuller test yielded a statistic of* -19.3198 *with a p-value of* 0.0000*, confirming the stationarity of the angular series. This property is fundamental for the validity of the econometric and machine learning models employed.*

5.3 Results of Predictive Models

Table 3 presents the comparative results of the six models evaluated on the test set (430 samples).

Table 3: Comparative Results of Models (NG=F)

Model	MAE(°)	Dir.Acc	R^2	MAPE(%)	Theil U	Bias(°)
AR(1)	2.1284	49.30%	-0.0003	100.66	0.6719	-0.0569
MLP	2.1270	51.40%	0.0011	101.13	0.6715	-0.0346
LSTM	2.1281	49.07%	-0.0004	99.98	0.6720	-0.0630
ARMA-GARCH	2.1448	50.70%	-0.0102	134.24	0.6753	—
ARMA-EGARCH	nan	0.00%	nan	nan	nan	—
ARMA-GJR-GARCH	2.1290	48.37%	-0.0006	100.56	0.6721	-0.0586

5.3.1 Analysis of Results

1. **MLP Model**: Achieves the best overall performance: lowest MAE (2.1270°), highest directional accuracy (51.40%), best R^2 (0.0011), and lowest bias (-0.0346°). The directional accuracy of 51.40% exceeds the random 50% level by 2.8%, demonstrating that the model captures useful predictive information.

2. **AR(1) Model**: The linear benchmark shows solid performance (MAE 2.1284°, DirAcc 49.30%), slightly inferior to MLP but competitive.

3. **LSTM Model**: Shows very similar results to AR(1) (MAE 2.1281°, DirAcc 49.07%), suggesting that long-term memory does not provide significant advantages for this particular series.

4. **GARCH Models**: The standard GARCH shows a higher MAE (2.1448°) and worse R^2 (-0.0102). EGARCH did not produce numerically stable predictions. GJR-GARCH, failing in its asymmetric specification, reverted to a simple ARMA alternative with results equivalent to AR(1).

6 Experimental Results: Brent Crude Oil (BZ=F)

6.1 Experimental Setup

To validate the methodology across different energy assets, we applied the exact same approach to Brent crude oil (BZ=F), another critical energy commodity highly sensitive

to geopolitical tensions. The study covers the same period: September 2017 to April 2026, totaling 2155 daily observations.

Table 4: Experimental Setup for Brent Crude Oil (BZ=F)

Parameter	**Value**
Asset	BZ=F (Brent Crude Oil, NYMEX)
Period	2017-09-10 to 2026-04-03
Frequency	Daily
Number of observations	2155
Calculated returns	2154
Return range	$[-0.2798, 0.1908]$
Angle range	$[-15.63°, 10.80°]$
Embedding window	$m = 6$ (determined by PACF)
Data split	1374 training, 344 validation, 430 test
Models evaluated	AR(1), MLP, LSTM, ARMA-GARCH, ARMA-EGARCH, ARMA-GJR-GARCH

6.2 Descriptive Statistics of the Angular Transformation (Brent)

Table 5 presents the descriptive statistics for Brent's angular series.

Table 5: Descriptive Statistics of Directional Angles (BZ=F)

Metric	**Value**
Arithmetic mean	0.0192°
Standard deviation	1.4588°
Minimum	−15.6296°
Maximum	10.8008°
Circular mean	0.0194°
Circular variance	0.000324
Circular standard deviation	0.0255 rad
95th percentile ($\|\theta\|$)	2.7482°

Remark 6.1 (Key Observation: Brent is Less Volatile)**.** *Compared to natural gas, Brent exhibits significantly lower angular volatility: a standard deviation of* 1.46° *versus* 2.60° *for natural gas. The 95th percentile of absolute angles is also much smaller (*2.75° *vs* 4.88°*). This suggests that Brent's directional dynamics are more stable and potentially more predictable than those of natural gas.*

Remark 6.2 (Stationarity for Brent)**.** *The augmented Dickey-Fuller test for Brent's angular series yielded a statistic of* -9.3268 *with a p-value of* 0.0000*, confirming stationarity (critical values:* $1\% = -3.4334$*,* $5\% = -2.8629$*,* $10\% = -2.5675$*).*

6.3 Results of Predictive Models (Brent)

Table 6 presents the comparative results of the models evaluated on Brent's test set (430 samples).

Table 6: Comparative Results of Models (BZ=F)

Model	MAE(°)	Dir.Acc	R^2	MAPE(%)	Theil U	Bias(°)
AR(1)	0.9455	51.86%	-0.0002	576580.49	0.6857	-0.0190
MLP	0.9466	50.47%	-0.0028	1447925.36	0.6865	-0.0095
LSTM	0.9456	51.86%	-0.0002	471584.00	0.6857	-0.0216
ARMA-GARCH	6.2907	51.86%	-20.8833	2.57e8%	3.2071	—
ARMA-EGARCH	nan	0.00%	nan	nan	nan	—
ARMA-GJR-GARCH	0.9455	51.86%	-0.0002	601128.76	0.6856	-0.0184

6.3.1 Analysis of Results (Brent)

1. **Surprise Finding**: Unlike natural gas, the simple AR(1) model matches or slightly outperforms the MLP and LSTM models for Brent. AR(1) achieves MAE of 0.9455° and directional accuracy of 51.86%.

2. **Directional Accuracy**: The best models achieve 51.86% directional accuracy, which exceeds the random 50% level by 3.7% — a meaningful edge in financial trading.

3. **GARCH Performance**: Standard GARCH fails dramatically for Brent (MAE 6.29°), confirming that GARCH models designed for raw returns do not transfer well to the angular space. GJR-GARCH (with ARMA fallback) performs equivalently to AR(1).

4. **Lower Errors**: The MAE for Brent (0.9455°) is substantially lower than for natural gas (2.1270°), consistent with Brent's lower angular volatility.

7 Comparative Analysis: Natural Gas vs. Brent

Table 7 presents a direct comparison between both energy assets.

Table 7: Comparative Analysis: Natural Gas vs. Brent Crude Oil

Characteristic	Natural Gas (NG=F)	Brent (BZ=F)
Angle range	$[-32.78°, 20.89°]$	$[-15.63°, 10.80°]$
Angular standard deviation	2.60°	1.46°
95th percentile ($\|\theta\|$)	4.88°	2.75°
Best model	MLP	AR(1) / LSTM
Best directional accuracy	51.40%	51.86%
Improvement over random	+2.8%	+3.7%
Best MAE	2.1270°	0.9455°

Theorem 7.1 (Asset-Specific Angular Signatures)**.** *Different energy assets exhibit distinct angular signatures. Natural gas shows higher angular dispersion (more erratic directional behavior), while Brent shows more compact angular distributions (more stable directionality). This implies that the optimal predictive model may be asset-dependent: MLP for volatile assets, simpler linear models for more stable ones.*

Remark 7.1 (Implications for Trading)**.** *The directional accuracy of approximately 52% achieved for both assets, while modest, is economically significant. In high-frequency trading or as a filter for existing strategies, a 2-4% edge above random can translate into consistent profitability when combined with proper risk management.*

8 Geopolitical Relevance: The Compass as a Navigation Tool

8.1 The Current Geopolitical Context (April 2026)

At the time of writing this paper, the world is witnessing an unprecedented escalation in tensions between the United States, Israel, and Iran. This conflict has profound implications for global energy markets, and particularly for natural gas and Brent crude oil — both highly sensitive to supply disruptions in the Middle East and strategic maritime routes.

Key elements of this conflict include:

- **The Strait of Hormuz**: Through which approximately 20% of the world's oil and a significant proportion of liquefied natural gas (LNG) transits. Iranian threats of a blockade have generated historic risk premiums.
- **Energy infrastructure**: Drone attacks and cyberattacks have affected production and storage facilities in the region.
- **Asymmetric volatility**: Geopolitical events tend to generate negative shocks (supply disruption, price increases) which, according to our results, produce pronounced leverage effects.

8.2 Why the Directional Compass is Relevant in Geopolitical Crises

Traditional models fail during crises for three fundamental reasons:

1. **Abrupt non-stationarity**: Parameters estimated during calm periods become invalid.
2. **Extreme heavy tails**: Geopolitical events generate price movements of 5-15% on individual days.
3. **Unexpected regime changes**: Market direction can reverse multiple times.

Theorem 8.1 (Stability of the Compass under Geopolitical Shocks)**.** *The transformation* $\theta_t = \arctan(r_t)$ *bounds the state space even under extreme movements:*

$$\lim_{|r_t| \to \infty} |\theta_t| = \frac{\pi}{2} \quad (90^\circ) \tag{19}$$

8.3 Regime Change Detection via the Turbulence Index

Definition 8.1 (Market Turbulence Index)**.** *We define the* ***turbulence index*** $\mathcal{T}_t$ *as the circular variance of the last* K *angles:*

$$\mathcal{T}_t = 1 - \frac{1}{K}\sqrt{\left(\sum_{i=0}^{K-1}\cos\theta_{t-i}\right)^2 + \left(\sum_{i=0}^{K-1}\sin\theta_{t-i}\right)^2} \tag{20}$$

Proposition 8.1 (Behavior of the Turbulence Index)**.**

- *Calm markets with a defined trend:* $\mathcal{T}_t \approx 0$
- *Sideways markets dominated by noise:* $\mathcal{T}_t \approx 0.5 - 0.7$
- *Geopolitical crises:* $\mathcal{T}_t$ *increases abruptly (leading indicator)*

8.4 Application to the Current Energy Context

Although our empirical study covered both natural gas and Brent, the methodology has immediate applicability to any energy asset sensitive to conflict. Table 8 presents an analysis of the expected response of the compass.

Table 8: Key Events and Expected Compass Response in Energy Markets

Event	**Expected Response**	**Implication**
Threat of closing the Strait of Hormuz	Angular dispersion ↑ (anticipation 3-5 days)	Early warning of crisis
Attack on energy infrastructure	Directional regime change	Hedging adjustment opportunity
Military response	Intensified leverage effect	Higher volatility post-negative shock
Diplomatic escalation	Reduction in circular variance	Possible return to normalcy

Operational Conclusion for April 2026:
In the current context of tensions between the United States, Israel, and Iran, the directional compass applied to both natural gas and Brent suggests special monitoring of the circular variance $\mathcal{T}_t$. The main advantage of the approach is its ability to operate in the bounded angular space, where even extreme price movements translate into manageable angles, preserving the numerical stability of the predictive model. Brent's lower angular volatility makes it potentially more predictable during crisis periods.

9 Discussion

9.1 Theoretical Implications

The results confirm the established theoretical properties for both energy assets:

- The transformation $\theta_t = \arctan(r_t)$ produces bounded angles for both assets (NG=F: $|\theta_t| \leq 32.78°$, BZ=F: $|\theta_t| \leq 15.63°$).
- The stationarity of the angular series (ADF: NG=F -19.32, BZ=F -9.33; both $p < 0.0001$) validates the use of models that assume this property.
- The circular correlation between predictions and actual values exceeds 0.999 for both assets, suggesting that the models adequately capture the general trend.

9.2 Model Comparison Across Assets

- **Natural Gas (volatile)**: MLP outperforms AR(1) and LSTM. Non-linearity adds value.
- **Brent (more stable)**: AR(1) matches or outperforms neural networks. Linearity suffices.
- **GARCH Models**: Consistently poor performance across both assets, confirming that volatility models designed for raw returns do not transfer to angular space.

9.3 Limitations of the Study

1. **Two energy assets only**: Validation on other asset classes (equities, currencies, cryptocurrencies) is required.
2. **Modest directional accuracy**: Although 51.86% exceeds random levels, there remains significant room for improvement.
3. **Inadequate loss function**: The use of MSE does not adequately penalize directional errors. Future work should implement circular loss functions.
4. **EGARCH instability**: The EGARCH model failed to produce stable predictions for both assets, indicating overparameterization.

10 Conclusions

10.1 Main Conclusions

This paper has presented a rigorous mathematical formalization of a geometric transformation for financial time series, emphasizing the novel interpretation as a **directional compass**, inspired by the Chinese invention that transformed maritime navigation. The main contributions are:

1. A vector representation $\vec{v}_t = (1, r_t)$ that induces an angular space $\theta_t = \arctan(r_t)$ with proven properties: Lipschitz continuity ($L = 1$), sign preservation, uniform boundedness, and scale invariance.
2. The introduction of the concept of a **financial directional compass** $\mathcal{C}(P_t, r_t) = (\theta_t, \rho_t)$ as a visualization tool, allowing the interpretation of price dynamics in terms of direction and normalized magnitude.

3. Empirical validation with 2155 observations of natural gas (NG=F, 2017-2026) and 2155 observations of Brent crude oil (BZ=F, 2017-2026), demonstrating that:
   - The angular series for both assets are stationary (ADF: -19.32 and -9.33, both $p < 0.0001$)
   - For natural gas, the MLP model achieves the best performance: MAE 2.1270°, DirAcc 51.40%
   - For Brent, the AR(1) and LSTM models achieve the best performance: MAE 0.9455°, DirAcc 51.86%
   - Directional accuracy exceeds random levels by 2.8% (natural gas) and 3.7% (Brent)
   - Brent exhibits lower angular volatility ($\sigma = 1.46°$ vs $2.60°$), suggesting more stable directional dynamics
4. A comparative analysis revealing that different assets have distinct "angular signatures", and the optimal predictive model may be asset-dependent.
5. An analysis of geopolitical relevance applied to the current conflict between the United States, Israel, and Iran (April 2026), demonstrating that the turbulence index $\mathcal{T}_t$ can act as a leading indicator of crises in energy markets.

10.2 The Compass as a Tool, not a Prophecy

It is important not to misinterpret the scope of this work. The proposed directional compass **does not predict the future** with certainty. Like the Chinese compass, it does not calm storms or create new seas. It simply offers a stable point of reference amidst uncertainty. Its value lies not in the accuracy of its point predictions — which for Brent achieve an MAE of 0.95° — but in its ability to:

- **Guide**: Indicate whether, on average, the market points towards positive or negative directions.
- **Warn**: Detect increases in circular variance that precede regime changes.
- **Stabilize**: Maintain numerical stability even when prices move erratically.

In the volatile geopolitical context of April 2026, where tensions between the United States, Israel, and Iran threaten to disrupt global energy supply, the directional compass offers exactly that: a tool to navigate, not to predict. Like those Chinese sailors who dared to cast off and trust a magnetized needle, investors and risk managers can use this compass to find their way through the increasingly turbulent seas of financial markets.

10.3 Future Work

1. **Circular loss function**: Implement $L_{\text{circ}}(\theta, \hat{\theta}) = 1 - \cos(\theta - \hat{\theta})$ to adequately penalize directional errors.
2. **Extension to other asset classes**: Equities (S&P500), currencies (EUR/USD), and cryptocurrencies (Bitcoin).

3. **Hybrid architectures**: Combine GARCH for volatility with MLP for direction.
4. **Real-time validation**: Implement a continuous monitoring system for the turbulence index $\mathcal{T}_t$.
5. **Multi-asset correlation**: Analyze angular correlations between different assets during crisis periods.

> *"Just as the Chinese compass did not create the sea but made it navigable, this financial compass does not eliminate market uncertainty, but offers a stable framework for finding one's way through it."*

A Supplementary Proofs

A.1 Lipschitz Inequality

Proof. For $r_i \leq r_j$:

$$|\arctan(r_j) - \arctan(r_i)| = \int_{r_i}^{r_j} \frac{dx}{1+x^2} \leq \int_{r_i}^{r_j} dx = r_j - r_i \tag{21}$$

□

A.2 Volatility-Angle Relationship

Proof. $\theta_t = \arctan(r_t) \Rightarrow r_t = \tan(\theta_t)$. Then:

$$\sigma^2_{\text{realized}} = \frac{1}{N-1} \sum_{t=1}^{N} \tan^2(\theta_t) \approx \frac{1}{N-1} \sum_{t=1}^{N} \theta_t^2 \tag{22}$$

for small angles. □

B Notation

- P_t: Price at time t
- r_t: Logarithmic return
- θ_t: Directional angle
- $\mathcal{T}_t$: Turbulence index
- NG=F: Natural Gas ticker on NYMEX
- BZ=F: Brent Crude Oil ticker on NYMEX

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